XRPL Validator — Daily Engineering Report · 2026-09-23

Period: 2026-09-23 00:00 to 24:00 US Eastern (2026-09-23 04:00 to 2026-09-24 04:00 UTC); ledgers #107,173,372 to #107,195,735. Every count from the ledger and the validator's logs stops at #107,195,735. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; signing records; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Normal (agreement not checked) — coverage outside planned maintenance 100.000% (objective 99.9%), agreement with the network not checked (the reference node no longer holds these ledgers' hashes), no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 21,835 of the 22,364 ledgers the network closed (97.635%) (agreement not checked: the ledgers' hashes are no longer held). Unsigned: 529 during planned restarts.
- **Native engine:** 5 engine receipts recorded (5 fixed, 0 explained, 0 without a recorded fix).
- **Changes:** 3 deployments (cycles 151, 152, 153), 20 commits merged into main, 40 engine findings fixed (F359–F398).

Service levels

Measure	Value
Ledgers closed by the network	22,364
Validations issued	21,835
Coverage (validations / ledgers closed)	97.635%
Validations matching the network's validated ledger hash	not checked: the reference node no longer holds these ledgers' hashes
Unsigned during planned restarts	529
Unsigned outside maintenance	0

Measure	Value
Coverage outside planned maintenance (objective 99.9%)	100.000%
Restarts	3
Basis: the hash the first peer announced as accepted (before cycle 183)	21,835

Restarts:

- 19:30:25 UTC: deployment of cycle 151; validations resumed at #107,187,940 (122 ledgers unsigned between the last validation before the restart and the first after it)
- 00:50:34 UTC: deployment of cycle 152; validations resumed at #107,192,890 (98 ledgers unsigned between the last validation before the restart and the first after it)
- 03:02:28 UTC: deployment of cycle 153; validations resumed at #107,195,151 (309 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,173,372–#107,195,735), by step:

Step	Ledgers with a receipt
Step 1 · native engine	5
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#107,190,696	Step 1 · native engine	1 transaction result(s) differed from rippled's	fixed: F397, commit 1e2d3d3, deployed in cycle 152 (2026-09-24 00:50 UTC)
R-02	—	#107,191,547	Step 1 · native engine	1 transaction result(s) differed from rippled's	fixed: F397 — NFTokenAcceptOffer checks the offer's Destination before funds; one account retried Destination-reserved NFT offers; deployed in cycle 152 (2026-09-24 00:50 UTC)
R-03	—	#107,191,549	Step 1 · native engine	1 transaction result(s) differed from rippled's	fixed: F397 — NFTokenAcceptOffer checks the offer's Destination before funds; one account retried Destination-reserved NFT offers; deployed in cycle 152 (2026-09-24 00:50 UTC)
R-04	—	#107,191,555	Step 1 · native	1 transaction result(s) differed from rippled's	fixed: F397 — NFTokenAcceptOffer checks the offer's Destination before funds; one account retried Destination-reserved NFT offers;

ID	Time	Ledger(s)	Step	Description	Disposition
			engine		deployed in cycle 152 (2026-09-24 00:50 UTC)
R-05	—	#107,194,228	Step 1 · native engine	4 object(s) differed in content	fixed: F398, commit ef17d8f, deployed in cycle 154 (2026-09-24 22:58 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-23-01, INC-01 is INC-2026-09-23-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

No incidents.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
151	`863ec8f`	19:30:25 UTC	n/a	n/a	402 matched, 0 mismatched	#26 from 19:57 UTC: ended early: superseded by the next deployment (2026-09-24 01:17 UTC)
152	`1e2d3d3`	00:50:34 UTC	n/a	n/a	404 matched, 0 mismatched	#27 from 01:17 UTC: ended early: superseded by the next deployment (2026-09-24 03:29 UTC)
153	`74c3b50`	03:02:28 UTC	n/a	n/a	411 matched, 0 mismatched	#28 from 03:29 UTC: ended early: superseded by the next deployment (2026-09-24 23:26 UTC)

Cycle 151 delivered: `c3c915c` feat(node): receipt canary — the live shadow plants a 1-drop diff on demand and proves its compare still flags it; `12a043f` fix(ledger): F359 — LedgerStateFix was never dispatched; port both fix types from rippled 3.4.0; `0e3f28b` fix(ledger): F360 — PermissionDelegationV1_1: DelegateSet, the delegate pays the fee, preFeeBalance_ is the account's own; `28aec17` fix(ledger): F361-F365 — AccountSet / SignerListSet rules from campaign 15 (first multi-signed specimens); `0a3dfb4` fix(ledger): F366-F369 — OracleSet time window and reserve, NFTokenModify order, noCurrency "1" (campaign 19); `dd90d91` test(node): F360 — campaign 22 (devnet PermissionDelegation): 29 delegation vectors + DelegateSet read-set; `fa5735e` fix(ledger): F370-F374 — MPT transfer-fee arithmetic through Number, legacy escrow divideRound, IssuanceSet/Clawback gates (campaign 18); `e2c806b` fix(ledger): F375-F380 — credential DepositAuth: MPT arm, XRP carve-out, AccountDelete, CredentialAccept/Delete order (campaign 20); `081001c` fix(ledger): F381 — OfferCreate preclaim's two tecFROZEN tests (campaign 17); `9010547` fix(ledger): F382 — a Batch's required signers come from each inner's initiator (the Delegate when present); `4789c7d` fix(ledger): F383-F388 — AMM deposit/withdraw sub-modes ported from rippled 3.4.0 (campaign 16); `74f5bbd` fix(ledger): F389-F394 — Batch inners: common preclaim, delegate permission, AccountTxnID order, AccountDelete fee, per-inner threading (campaign 23); `e142bc1` test(node): campaign 23 — 60 byte-exact Batch × feature vectors; fetcher read-sets, build_batch.py; `a05e2dc` test(node): probe_bundle judges a Batch whole — per-inner threading and inner verdicts; `ba89585` test(node): campaign 24 — Batch × NFT, offer, check, channel, escrow, oracle, DID, AMM, trust line, AccountSet (27 vectors); `f7ad693` fix(ledger): F395-F396 — Batch

touched sets read decoded pre-images; inner ids use the shared canonicaliser; `7857d61` docs: README — live-shadow status and progress history; `863ec8f` docs: README — second clean 24-hour run; link the daily reports

Cycle 152 delivered: `1e2d3d3` fix(ledger): F397 — NFTokenAcceptOffer preclaim in rippled's order: Destination before funds (soak #26 receipt)

Cycle 153 delivered: `74c3b50` fix(node): UNL from unl.xrplf.org, retried until it loads; watch_engine counts canary lines apart

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
04:02 UTC	xrpl-validator	`9010547`	fix(ledger): F382 — a Batch's required signers come from each inner's initiator (the Delegate when present)
04:02 UTC	xrpl-validator	`081001c`	fix(ledger): F381 — OfferCreate preclaim's two tecFROZEN tests (campaign 17)
04:02 UTC	xrpl-validator	`e2c806b`	fix(ledger): F375-F380 — credential DepositAuth: MPT arm, XRP carve-out, AccountDelete, CredentialAccept/Delete order (campaign 20)
04:02 UTC	xrpl-validator	`fa5735e`	fix(ledger): F370-F374 — MPT transfer-fee arithmetic through Number, legacy escrow divideRound, IssuanceSet/Clawback gates (campaign 18)
04:02 UTC	xrpl-validator	`dd90d91`	test(node): F360 — campaign 22 (devnet PermissionDelegation): 29 delegation vectors + DelegateSet read-set
04:02 UTC	xrpl-validator	`0a3dfb4`	fix(ledger): F366-F369 — OracleSet time window and reserve, NFTokenModify order, noCurrency "1" (campaign 19)
04:02 UTC	xrpl-validator	`28aec17`	fix(ledger): F361-F365 — AccountSet / SignerListSet rules from campaign 15 (first multi-signed specimens)
04:02 UTC	xrpl-validator	`0e3f28b`	fix(ledger): F360 — PermissionDelegationV1_1: DelegateSet, the delegate pays the fee, preFeeBalance_ is the account's own
04:02 UTC	xrpl-validator	`12a043f`	fix(ledger): F359 — LedgerStateFix was never dispatched; port both fix types from rippled 3.4.0
05:56 UTC	xrpl-validator	`4789c7d`	fix(ledger): F383-F388 — AMM deposit/withdraw sub-modes ported from rippled 3.4.0 (campaign 16)
07:07 UTC	xrpl-validator	`e142bc1`	test(node): campaign 23 — 60 byte-exact Batch × feature vectors; fetcher read-sets, build_batch.py
07:07 UTC	xrpl-validator	`74f5bbd`	fix(ledger): F389-F394 — Batch inners: common preclaim, delegate permission, AccountTxnID order, AccountDelete fee, per-inner threading (campaign 23)
08:10 UTC	xrpl-validator	`a05e2dc`	test(node): probe_bundle judges a Batch whole — per-inner threading and inner verdicts
08:10 UTC	xrpl-validator	`ba89585`	test(node): campaign 24 — Batch × NFT, offer, check, channel, escrow, oracle, DID, AMM, trust line, AccountSet (27 vectors)
08:28 UTC	xrpl-validator	`f7ad693`	fix(ledger): F395-F396 — Batch touched sets read decoded pre-images; inner ids use the shared canonicaliser
14:42 UTC	xrpl-validator	`7857d61`	docs: README — live-shadow status and progress history
16:25 UTC	xrpl-validator	`863ec8f`	docs: README — second clean 24-hour run; link the daily reports
23:05 UTC	xrpl-validator	`1e2d3d3`	fix(ledger): F397 — NFTokenAcceptOffer preclaim in rippled's order: Destination before funds (soak #26 receipt)

Merged	Repo	Commit	Subject
02:08 UTC	xrpl-validator	`74c3b50`	fix(node): UNL from unl.xrplf.org, retried until it loads; watch_engine counts canary lines apart
03:30 UTC	xrpl-validator	`ef17d8f`	fix(ledger): F398 — OfferStream original funds read through the afView's balance hooks (soak #27 receipt)

Findings fixed

- **fix(ledger): F382 — a Batch's required signers come from each inner's initiator (the Delegate when present)** (`9010547` , xrpl-validator) rippled's Batch::preflightSigValidated builds the required batch signers from rb.getInitiator() — the inner's sfDelegate when it carries one, because the delegate signed it — plus an sfCounterparty and a co-signing sfSponsor, each unless it is the outer account (Batch.cpp:423-445). We used the inner's Account, so a validated batch with a delegated inner would be refused here as temBAD_SIGNER. Reachable once …
- **fix(ledger): F381 — OfferCreate preclaim's two tecFROZEN tests (campaign 17)** (`081001c` , xrpl-validator) Campaign 17 (testnet, 164 landed: the OfferCreate flag matrix — Passive, IoC, FoK (fixFillOrKill), Sell, TickSize, TransferRate, self-cross, expiry, OfferSequence, unfunded/frozen makers, reserve edges, freezes) — 156/164 on the live engine; 7 misses were one rule, in two places: - checkGlobalFrozen on the TakerPays asset, then TakerGets, right after the account and ahead of funding and expiry: an issuer under …
- **fix(ledger): F375-F380 — credential DepositAuth: MPT arm, XRP carve-out, AccountDelete, CredentialAccept/Delete order (campaign 20)** (`e2c806b` , xrpl-validator) Campaign 20 (testnet, 117 landed: DepositAuth by credentials across Payment, EscrowFinish, PaymentChannelClaim, CheckCash, AccountDelete) — 11 misses, six rules: - F375: the MPT payment arm runs the whole verifyDepositPreauth (expired cleanup, credential-set authorization), not only the by-account preauth. - F376: on a direct XRP payment verifyDepositPreauth — cleanup included — runs only when the amount or the …
- **fix(ledger): F370-F374 — MPT transfer-fee arithmetic through Number, legacy escrow divideRound, IssuanceSet/Clawback gates (campaign 18)** (`fa5735e` , xrpl-validator) Campaign 18 (testnet, 124 landed MPT txs) — 15 misses, four rules, plus one Batch-era rule found reading the same code: - F370: MPT direct payments with a TransferFee. rippled's "No rounding" means no rounding MODE: the cost is multiply(amount, rate) = Number product (16-digit mantissa, to nearest) cast to int64 (to nearest, ties even); a partial delivery is divide(SendMax, rate) = muldiv(num, 1e17, den) + 5 then to …
- **fix(ledger): F366-F369 — OracleSet time window and reserve, NFTokenModify order, noCurrency "1" (campaign 19)** (`0a3dfb4` , xrpl-validator) Campaign 19 (testnet, 114 landed: NFTokenModify, NFTokenCancelOffer, Oracle, DID depth) — 10 misses, four rules: - F366: OracleSet's ±300 s LastUpdateTime window is centred on the header closeTime of the ledger being BUILT = parent close + closeTimeResolution, not the parent close every hasExpired test uses (3-25 tec / 3-26 tes). - F367: OracleSet tests the reserve for every adjustment (+2..-1) against …
- **fix(ledger): F361-F365 — AccountSet / SignerListSet rules from campaign 15 (first multi-signed specimens)** (`28aec17` , xrpl-validator) Campaign 15 (testnet, 138 landed txs: AccountSet depth, SetRegularKey, SignerListSet, 13 multi-signed) — 130/138 byte-exact on the live engine; the 8 misses were five rules: - F361: an unknown SetFlag/ClearFlag (32, 99) is a silent no-op; the `>= 32 → temMALFORMED` rule had no specimen behind it. - F362: the AccountTxnID prior check reads an absent field as the zero hash (getFieldH256 on a non-present optional), not …

- **fix(ledger): F360 — PermissionDelegationV1_1: DelegateSet, the delegate pays the fee, preFeeBalance_ is the account's own** (`0e3f28b`, xrpl-validator) PermissionDelegationV1_1 reached majority on mainnet 2026-09-21 (active ~10-05); the engine had no DelegateSet and no delegated fee path. rippled 3.4.0: - Transactor::apply consumes the ACCOUNT's sequence/ticket but payFee charges the fee payer = getInitiator() = sfDelegate when present. apply_common now does the same (TxFields::delegate()). - preFeeBalance_ is the account's balance before the fee. Every site that …
- **fix(ledger): F359 — LedgerStateFix was never dispatched; port both fix types from rippled 3.4.0** (`12a043f`, xrpl-validator) fixNFTokenPageLinks and fixCleanup3_2_0 are enabled on mainnet, so any mainnet LedgerStateFix would have applied as tecUNSUPPORTED against the network's code — a guaranteed receipt. Found by checking the 3.4.0 transaction list against dispatch.rs.
- **fix(ledger): F383-F388 — AMM deposit/withdraw sub-modes ported from rippled 3.4.0 (campaign 16)** (`4789c7d`, xrpl-validator) Testnet campaign 16 (AMM deposit/withdraw sub-modes against rippled 3.4.0) found six rules the engine lacked or got wrong: 16 of 138 bundles failed under BOTH engines (the transactors are shared). Each fix is a structural port — rippled's function body line for line, in its operation order and Number rounding modes, with its result codes in its order — on the signed Number port (tx::number) where the unsigned n_* …
- **fix(ledger): F389-F394 — Batch inners: common preclaim, delegate permission, AccountTxnID order, AccountDelete fee, per-inner threading (campaign 23)** (`74f5bbd`, xrpl-validator) Campaign 23 (devnet, 2026-09-23) crossed Batch (BatchV1_1 activates on mainnet ~09-29) with delegation, MPT, credentials, tickets, AccountDelete and LedgerStateFix inners. Six rules the engine had wrong: F389 (6-2, 6-5, 6-6, 6-7, 6-9) — a Batch inner runs rippled's whole per-transaction pipeline, common preclaim included (checkSeqProxy, checkPriorTxAndLastLedger — applySteps.cpp:179-183): an inner with a future/past …
- **fix(ledger): F395-F396 — Batch touched sets read decoded pre-images; inner ids use the shared canonicaliser** (`f7ad693`, xrpl-validator) A fresh-context review of F389-F394 against rippled 3.4.0 found two real problems (no critical one in the live-shadow path): F395 — tx/batch.rs built each inner's touched set with the pre-batch value from `state_map.lookup`, the RAW leaf. state_replay stores its leaves binary behind a `leaf_decoder`, so binary was compared with the sandbox's JSON: every unchanged write-back of a base object looked like a change (a …
- **fix(ledger): F397 — NFTokenAcceptOffer preclaim in rippled's order: Destination before funds (soak #26 receipt)** (`1e2d3d3`, xrpl-validator) Soak #26 receipt, mainnet #107190696 072B81D6348C: rKFaUach accepted sell offer C4D78B6A, whose Destination is rpx9JThQ, while also short of the 7 XRP price. Mainnet: tecNO_PERMISSION. Ours: tecINSUFFICIENT_FUNDS. The bundle probe reproduced it on the live build.
- **fix(ledger): F398 — OfferStream original funds read through the afView's balance hooks (soak #27 receipt)** (`ef17d8f`, xrpl-validator) rippled's strandFlow opens `PaymentSandbox afView(&baseView)` beside the strand's own sandbox (StrandFlow.h:133), and TOfferStreamBase judges an unfunded or tiny offer "found" (permanently removed) only when `accountFundsHelper(cancelView_ = afView)` equals `ownerFunds_`. Both reads end in the PaymentSandbox hooks — `balanceHookIOU` (IOU lines and `xrpLiquid`'s full balance) and `ownerCountHook` — which walk the …

Other engineering changes

- **fix(node): UNL from unl.xrplf.org, retried until it loads; watch_engine counts canary lines apart** (`74c3b50`, xrpl-validator) Since 2026-09-21 every live_viewer start came up with the consensus monitor frozen at

establish(round=1), UNL 0: the one startup fetch of the validator list failed on both sources, and the consensus loader answered "consensus disabled" for the life of the process. Both failures are TLS, both on the publishers' side: - vl.ripple.com, since its 2026-09-21 certificate renewal, serves the leaf (CN=*.ripple.com) twice …

Tests and campaigns

- **test(node): F360 — campaign 22 (devnet PermissionDelegation): 29 delegation vectors + DelegateSet read-set** (`dd90d91` , xrpl-validator)
- **test(node): campaign 23 — 60 byte-exact Batch × feature vectors; fetcher read-sets, build_batch.py** (`e142bc1` , xrpl-validator)
- **test(node): probe_bundle judges a Batch whole — per-inner threading and inner verdicts** (`a05e2dc` , xrpl-validator)
- **test(node): campaign 24 — Batch × NFT, offer, check, channel, escrow, oracle, DID, AMM, trust line, AccountSet (27 vectors)** (`ba89585` , xrpl-validator)

Docs and operations

- **docs: README — live-shadow status and progress history** (`7857d61` , xrpl-validator)
- **docs: README — second clean 24-hour run; link the daily reports** (`863ec8f` , xrpl-validator)

Independence program

No independence record for UTC day 2026-09-23; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Voting events:

- PermissionDelegationV1_1 lost majority — ledger 107,181,569, 2026-09-23 12:47 UTC

Amendment	Majority since	Activation due
BatchV1_1	2026-09-15 14:06 UTC	2026-09-29 14:06 UTC

Amendments enabled on mainnet: 94.

Open items

None.

Notes

Hand-written notes on the day's XRPL work: research and operations.

- **PermissionDelegationV1_1 lost majority on mainnet.** The network applied `EnableAmendment` with `tfLostMajority` at ledger 107,181,569 (2026-09-23 12:47 UTC). It had held a majority since 2026-09-21 and would have activated around 2026-10-05; the two-week clock restarts only if it regains majority. **BatchV1_1 still holds its majority** (since 2026-09-15 14:06 UTC) and activates around 2026-09-29 14:06 UTC.
- **The second clean 24-hour run.** The live soak started 2026-09-22 11:38 EDT passed its 24-hour mark with zero engine receipts — every compared ledger matched rippled byte-for-byte (the first such run was 2026-09-20/21).
- **Batch conformance ahead of activation.** Campaign 23 (devnet) crossed Batch with delegation, MPT, credentials, tickets, AccountDelete and LedgerStateFix inners: six engine rules fixed (F389–F394). The subtlest: rippled threads an owner's AccountRoot (`threadOwners`) only for nodes an inner itself inserted or erased, in that inner's own ApplyStateTable — not for whatever the batch as a whole created. Campaign 24 (devnet) put NFT, offer, check, payment-channel, escrow, oracle, DID, AMM, trust-line and AccountSet inners inside Batches, each shaped so a later inner uses what an earlier one created: 27/27 byte-exact on the first run.
- **An independent review against the rippled 3.4.0 sources** of the Batch fixes found two more issues, fixed test-first (F395, F396): a replay-only pre-image read of raw leaves, and inner transaction ids hashed without the codec's canonical spelling of decimal MPT amounts and short-hex `AssetPrice`.
- **Desk checks for the two Batch-only bug classes found so far:** no other transactor copies raw transaction `Flags` into a ledger object (where `tfInnerBatchTxn` would leak), and none gates the fee level in preclaim (a Batch inner carries `Fee: "0"`).
- **Deployment.** The engine build f7ad693 (findings through F396) replaced the live shadow build at 15:31 EDT after its gate (1,589 suite tests and 706 flow-engine tests passing) and two mainnet replay windows (60 and 98 ledgers, zero mismatches). The post-deployment health check matched 402 ledgers with no mismatches, and a new 24-hour soak began at 15:57 EDT (seed ledger 107,187,825).
- **Receipt canary — first live run.** One drop was planted in the shadow's copy of a mainnet AccountRoot at ledger 107,188,304; the byte-for-byte compare flagged it and the receipt watcher logged it 255 seconds later. Canary lines are never counted as receipts: the check exists to prove that a zero-receipt soak means the comparison was running and sensitive. It now runs daily.
- **Soak #26 receipt and finding F397.** At ledger 107,190,696 an NFTokenAcceptOffer was submitted by an account other than the offer's Destination, which was also short of the 7 XRP price. Mainnet returned `tecNO_PERMISSION`; the engine returned `tecINSUFFICIENT_FUNDS` — a fee-only failure either way. rippled judges the Destination before the funds; the engine's accept-offer preclaim now follows rippled 3.4.0's order exactly (F397, with the mainnet transaction pinned as a regression vector). Verified and merged; deployment awaits a decision.
- **Second deployment (F397).** Build 1e2d3d3 replaced the live shadow build at 20:51 EDT after its gate (1,594 suite tests and 706 flow-engine tests passing) and both mainnet replay windows (zero mismatches). The health

check matched 404 ledgers with no mismatches, and soak #27 began at 21:17 EDT (seed ledger 107,192,792). Soak #26's four receipts — all the same NFTokenAcceptOffer case, from one account retrying offers reserved for another — replay byte-exact on the fixed build and were archived.

- **Soak #27 receipt and finding F398.** At ledger 107,194,228 an immediate-or-cancel OfferCreate sold 942.58 XRP for RLUSD across one maker's five offers at a single quality. The payment engine's first pass consumed four of them and left the fifth backed by about 10^{-14} RLUSD; on the second pass rippled found that offer too small to trade and removed it. rippled tells an offer *found* unfunded or tiny (removed) from one that *became* so during the payment (skipped) by comparing the owner's funds with the funds read through the strand's "all funds" view, whose balance reads apply the payment's deferred-credit table. The engine read that view's raw trust-line balance (1.5×10^{-14} against the adjusted 1.0×10^{-14}), judged the offer to have become tiny, and kept it. Both original-funds reads (trust line and XRP) now apply the same adjustments as rippled 3.4.0's `afView` (F398), with the mainnet transaction pinned as a regression vector. Verified and merged; deployment awaits a decision.
- **Third deployment (validator-list sources).** Build 74c3b50 replaced the live shadow build at 23:03 EDT after its gate (1,596 suite tests and 706 flow-engine tests passing) and both mainnet replay windows (60 and 98 ledgers, zero mismatches). The node now fetches its validator list from unl.xrplf.org as well as vl.ripple.com, verifies the publisher's signature, and retries with backoff when no list can be fetched instead of running without consensus tracking. The list loaded from unl.xrplf.org (35 validators) and the consensus monitor resumed. The health check matched 411 ledgers with no mismatches, and soak #28 began at 23:29 EDT (seed ledger 107,194,842). Soak #27 ended at the deployment with the single receipt above.
- **Deployment practice.** The node's start-up state sync (19.9 million ledger objects, read from the reference rippled node) took 19.4 minutes instead of the usual 5.5–7, because a test build was running on the reference node's machine at the same time. Builds and replays there are now held automatically until a deployment's health check completes.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Hub-Mediated Multilateral Check Netting** (settlement) Add transaction ttCHECK_NET carrying sfCheckIDs (STVector256, bounded ≤8) referencing existing Check objects where the submitting account is either sf
- **On-Ledger Settlement Finality Receipts** (settlement) Add an optional tfFinalityReceipt flag to EscrowFinish and CheckCash; when set and the discharge succeeds, doApply additionally creates a compact ltSE
- **Destination-Enforced Travel-Rule Envelope** (compliance) Add lsfRequireTravelRule (AccountSet-settable flag) and sfTravelRuleThreshold to AccountRoot, plus a new bounded VL SField sfTravelRuleEnvelope (max 4

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-23 03:49	09-23 04:02	xrpl-validator	`9010547`	fix(ledger): F382 — a Batch's required signers come from each inner's initiator (the Delegate when present)
09-23 03:33	09-23 04:02	xrpl-validator	`081001c`	fix(ledger): F381 — OfferCreate preclaim's two tecFROZEN tests (campaign 17)
09-23 03:30	09-23 04:02	xrpl-validator	`e2c806b`	fix(ledger): F375-F380 — credential DepositAuth: MPT arm, XRP carve-out, AccountDelete, CredentialAccept/Delete order (campaign 20)
09-23 03:23	09-23 04:02	xrpl-validator	`fa5735e`	fix(ledger): F370-F374 — MPT transfer-fee arithmetic through Number, legacy escrow divideRound, IssuanceSet/Clawback gates (campaign 18)
09-23 03:14	09-23 04:02	xrpl-validator	`dd90d91`	test(node): F360 — campaign 22 (devnet PermissionDelegation): 29 delegation vectors + DelegateSet read-set
09-23 03:06	09-23 04:02	xrpl-validator	`0a3dfb4`	fix(ledger): F366-F369 — OracleSet time window and reserve, NFTokenModify order, noCurrency "1" (campaign 19)
09-23 03:00	09-23 04:02	xrpl-validator	`28aec17`	fix(ledger): F361-F365 — AccountSet / SignerListSet rules from campaign 15 (first multi-signed specimens)
09-23 02:53	09-23 04:02	xrpl-validator	`0e3f28b`	fix(ledger): F360 — PermissionDelegationV1_1: DelegateSet, the delegate pays the fee, preFeeBalance_ is the account's own
09-23 02:43	09-23 04:02	xrpl-validator	`12a043f`	fix(ledger): F359 — LedgerStateFix was never dispatched; port both fix types from rippled 3.4.0
09-23 05:54	09-23 05:56	xrpl-validator	`4789c7d`	fix(ledger): F383-F388 — AMM deposit/withdraw sub-modes ported from rippled 3.4.0 (campaign 16)
09-23 06:57	09-23 07:07	xrpl-validator	`e142bc1`	test(node): campaign 23 — 60 byte-exact Batch × feature vectors; fetcher read-sets, build_batch.py
09-23 06:57	09-23 07:07	xrpl-validator	`74f5bbd`	fix(ledger): F389-F394 — Batch inners: common preclaim, delegate permission, AccountTxnID order, AccountDelete fee, per-inner threading (campaign 23)
09-23 07:15	09-23 08:10	xrpl-validator	`a05e2dc`	test(node): probe_bundle judges a Batch whole — per-inner threading and inner verdicts
09-23 07:47	09-23 08:10	xrpl-validator	`ba89585`	test(node): campaign 24 — Batch × NFT, offer, check, channel, escrow, oracle, DID, AMM, trust line, AccountSet (27 vectors)
09-23 08:21	09-23 08:28	xrpl-validator	`f7ad693`	fix(ledger): F395-F396 — Batch touched sets read decoded pre-images; inner ids use the shared canonicaliser
09-23 14:42	09-23 14:42	xrpl-validator	`7857d61`	docs: README — live-shadow status and progress history
09-23 16:25	09-23 16:25	xrpl-validator	`863ec8f`	docs: README — second clean 24-hour run; link the daily reports
09-23 23:05	09-23 23:05	xrpl-validator	`1e2d3d3`	fix(ledger): F397 — NFTokenAcceptOffer preclaim in rippled's order: Destination before funds (soak #26 receipt)
09-24 02:08	09-24 02:08	xrpl-validator	`74c3b50`	fix(node): UNL from unl.xrplf.org, retried until it loads; watch_engine counts canary lines apart
09-24 03:30	09-24 03:30	xrpl-validator	`ef17d8f`	fix(ledger): F398 — OfferStream original funds read through the afView's balance hooks (soak #27 receipt)

Times in UTC. A commit appears on the day it was committed to main or merged into it.